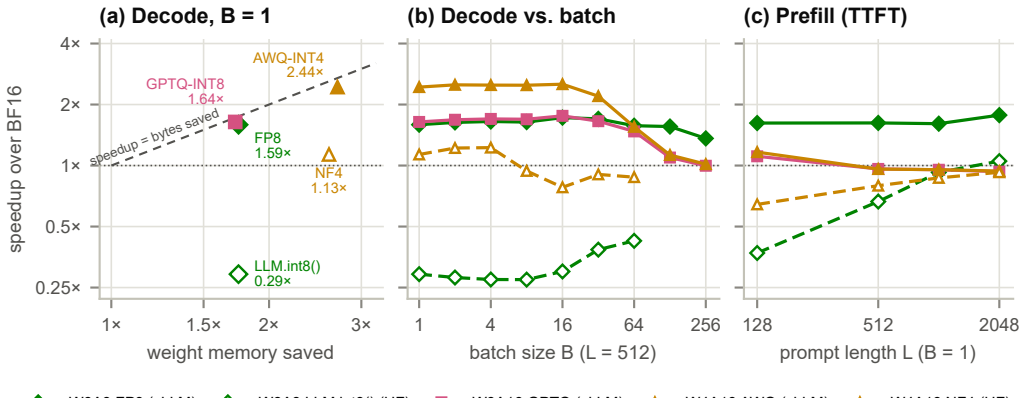
(a) Decode, B = 1
(b) Decode vs. batch
(c) Prefill (TTFT)
speedup over BF16
4×
2×
1×
0.5×
0.25×
AWQ-INT4
2.44×
GPTQ-INT8
1.64×
speedup = bytes saved
FP8
1.59×
NF4
1.13×
LLM.int8()
0.29×
1×
1.5×
2×
3×
weight memory saved
1
4
16
64
256
batch size B (L = 512)
128
512
2048
prompt length L (B = 1)
W8A8 FP8 (vLLM)
W8A8 LLM.int8() (HF)
W8A16 GPTQ (vLLM)
W4A16 AWQ (vLLM)
W4A16 NF4 (HF)